Moving Points – Method of Polynomials and Parametric Functions

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This material was developed by Eric Ransom, a former Mexican IMO contestant and silver medalist, who serves as the academic lead for geometry in the training programs of various international olympiad teams. He is one of the best current trainers in Mexico and undoubtedly among the top international instructors in the field of geometry. Thanks to his work, the development of geometry in Mexico continues to grow and remains a reference point for the olympiad community.

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1 Definitions

1.1 Parametric and Rational Functions

Definition 1.1 (Parametric Function). *Let $f_1, f_2, \ldots, f_n$ be functions $f_i : \mathbb{R} \to \mathbb{R}$ for $i = \{1, 2, \ldots, n\}$. A parametric function $F : \mathbb{R} \to \mathbb{R}^n$ is a function with a parameter $t \in \mathbb{R}$ such that*

$$F(t) = (f_1(t), f_2(t), \ldots, f_n(t)).$$

Definition 1.2 (Rational Function). *Let P and Q be polynomials. A function $f : \mathbb{R} \to \mathbb{R}$ is rational if it is of the form*

$$f(x) = \frac{P(x)}{Q(x)}.$$

Example 1.3 (Rational parametrization of the unit circle). *If $F : \mathbb{R} \cup \{\infty\} \to \mathbb{R}^2$ is such that*

$$F(t) = \left(\frac{2t}{t^2+1}, \frac{t^2-1}{t^2+1}\right),$$

and the image of $t = \infty$ is defined by continuity, the geometric locus of F is the unit circle.

Proof. We want to verify that

$$\left(\frac{2t}{t^2+1}\right)^2 + \left(\frac{t^2-1}{t^2+1}\right)^2 = 1.$$

This is verified directly. Showing that for each point on the unit circle there exists a t is left as an exercise.

1.2 Line Coordinates

Definition 1.4 (Line Coordinates). *Let ℓ be a line that does not pass through the origin. If ℓ is given by $ax + by + 1 = 0$, then we write $\ell = [a, b]$.*

Example 1.5 (Rational parametrization of a line tangent to the unit circle). *If $F : \mathbb{R} \cup \{\infty\} \to \mathbb{R}^2$ is such that*

$$F(t) = \left[\frac{2t}{t^2+1}, \frac{t^2-1}{t^2+1}\right],$$

and the image of $t = \infty$ is defined by continuity, then the geometric locus of F is the set of lines tangent to the unit circle.

1.3 Homogeneous Coordinates

Definition 1.6 (Real Projective Plane). *The set of all lines that pass through the origin in $\mathbb{R}^3$ is called the real projective plane $\mathbb{RP}^2$.*

Definition 1.7 (Homogeneous Coordinates – Points). *In $\mathbb{RP}^2$, a point P has coordinates $(a : b : c) = (\lambda a : \lambda b : \lambda c)$, with a, b, c not all zero. The point P represents the line that passes through $(0, 0, 0)$ and (a, b, c) in $\mathbb{R}^3$.*

Definition 1.8 (Homogeneous Coordinates – Lines). *In $\mathbb{RP}^2$, a line ℓ has coordinates $[a : b : c] = [\lambda a : \lambda b : \lambda c]$, representing the plane $ax + by + cz = 0$ in $\mathbb{R}^3$.*

Theorem 1.9 (Dot Product – Incidence). *A point $P = (x : y : z)$ lies on the line $\ell = [a : b : c]$ if and only if $ax + by + cz = 0$.*

Theorem 1.10 (Cross Product – Intersection). *For real numbers a_i, b_i, c_i with $i = 1, 2$ the following holds:*

- *The intersection of two lines $\ell_1 = [a_1 : b_1 : c_1]$ and $\ell_2 = [a_2 : b_2 : c_2]$ is*

$$P = (b_1c_2 - c_1b_2 : c_1a_2 - a_1c_2 : a_1b_2 - b_1a_2).$$

- *The line that passes through two points $P_1 = (a_1 : b_1 : c_1)$ and $P_2 = (a_2 : b_2 : c_2)$ is*

$$\ell = [b_1c_2 - c_1b_2 : c_1a_2 - a_1c_2 : a_1b_2 - b_1a_2].$$

1.4 Moving Point and Moving Line

Definition 1.11 (Moving Point). *Let P, Q, R be polynomials with no common roots. A parametric function $F : \mathbb{R} \cup \{\infty\} \to \mathbb{RP}^2$ of the form*

$$F(t) = (P(t) : Q(t) : R(t))$$

is called a moving point. Its degree is

$$\deg(F) = \max\{\deg(P), \deg(Q), \deg(R)\}.$$

Definition 1.12 (Moving Line). *Analogously, a function $F : \mathbb{R} \cup \{\infty\} \to \mathbb{RP}^2$ of the form*

$$F(t) = [P(t) : Q(t) : R(t)]$$

is a moving line of degree

$$\deg(F) = \max\{\deg(P), \deg(Q), \deg(R)\}.$$

2 Computing Degrees

Theorem 2.1 (Projective map between domain and range of the same type). *Let $f : C_1 \to C_2$ be a projective map. If F is a moving point or line, then $\deg(F) = \deg(f(F))$.*

Theorem 2.2 (Zack's Lemma). *The following holds:*

- *If A and B are moving points of degree a and b with $A = B$ in k cases, then the line AB has degree $\leq a + b - k$.*
- *If ℓ_A and ℓ_B are moving lines with $\ell_A = \ell_B$ in k cases, then their intersection has degree $\leq a + b - k$.*

Theorem 2.3 (Projective map: line to conic and vice versa). *The following holds:*

- *C_1 line, C_2 conic: $2 \cdot \deg(F) = \deg(f(F))$.*
- *C_1 conic, C_2 line: $\frac{1}{2} \cdot \deg(F) = \deg(f(F))$.*

Theorem 2.4 (Vladyslav's Lemma). *If A and B are moving points on a conic with $\deg(A) = \deg(B)$, then $\deg(AB) = \deg(A) = \deg(B)$.*

Theorem 2.5 (Polar Transformation). *The polar transformation preserves the degree.*

3 Concluding

Theorem 3.1 (Collinearity – Concurrence). *The following holds:*

- *If A, B, C are moving points of degrees a, b, c and are collinear in $a + b + c + 1$ cases, then they are always collinear.*
- *If ℓ_A, ℓ_B, ℓ_C are moving lines concurrent in $a+b+c+1$ cases, then they are always concurrent.*

Theorem 3.2 (Equal points). *If $A = B$ in $a + b + 1$ cases, then $A = B$ always.*

Theorem 3.3 (Degree 1). *A moving point of degree 1 traces a line; a moving line of degree 1 generates a pencil of lines.*

Theorem 3.4 (Degree 2). *A moving point of degree 2 traces a conic or collinear points. Moving lines of degree 2 are tangent to a conic or concurrent.*

Theorem 3.5 (Cyclic quadrilateral). *A quadrilateral $ABCD$ with moving points of degrees a, b, c, d is cyclic if it is so in $\leq 2(a+b+c+d)+1$ cases.*

4 Examples

Problem 4.1 (Kariya's Theorem). *Let ABC be a triangle with incenter I and intouch triangle DEF. Select X, Y, Z on rays ID, IE, IF, respectively, such that $IX = IY = IZ$. Show that lines AX, BY, CZ concur.*

Problem 4.2 (IGO 2015 P3). *Let H be the orthocenter of the triangle ABC, and pass two lines ℓ_1 and ℓ_2 through H such that $\ell_1 \perp \ell_2$. We have $\ell_1 \cap BC = D$ and $\ell_1 \cap AB = Z$, and also $\ell_2 \cap BC = E$ and $\ell_2 \cap AC = X$. Pass a line d_1 through D parallel to AC and another line d_2 through E parallel to AB, and let $d_1 \cap d_2 = Y$. Prove X, Y and Z are collinear.*

Problem 4.3 (USA TST 2015 P6). *Let ABC be a non-equilateral triangle and let M_a, M_b, M_c be the midpoints of the sides BC, CA, AB, respectively. Let S be a point lying on the Euler line. Denote by X, Y, Z the second intersections of M_aS, M_bS, M_cS with the nine-point circle. Prove that AX, BY, CZ are concurrent.*

5 Problems

Problem 5.1 (Kiepert's Theorem). *Let ABC be a triangle and select points X, Y, Z such that $\triangle BXC$, $\triangle AYC$, $\triangle AZB$ are similar triangles with $BX = XC$, $CY = YA$, $AZ = ZB$. Show that lines AX, BY, CZ concur.*

Problem 5.2 (Rogelio Guerrero Original). *Let ABC be a triangle with circumcenter O and let C' be the reflection of C over AB. Let $D = CO \cap AB$, $E = AC \cap (\circ CBC')$, and $F = BC' \cap (\circ CAC')$. Prove that D, E and F are collinear.*

Problem 5.3 (RMM 2016 P1). *Let ABC be a triangle and let D be a point on the segment BC, $D \neq B$ and $D \neq C$. The circle ABD meets the segment AC again at an interior point E. The circle ACD meets the segment AB again at an interior point F. Let A' be the reflection of A in the line BC. The lines $A'C$ and DE meet at P, and the lines $A'B$ and DF meet at Q. Prove that the lines AD, BP and CQ are concurrent (or all parallel).*

Problem 5.4 (Iran TST 2005 P2). *Assume ABC is an isosceles triangle with $AB = AC$. Suppose P is a point on the extension of side BC. Let X and Y be points on AB and AC such that $PX \parallel AC$ and $PY \parallel AB$. Let T be the midpoint of arc BC that does not include A. Prove that $PT \perp XY$.*

Problem 5.5 (Cyberspace Math Competition 2020 P3). *Let ABC be a triangle such that $AB > BC$ and let D be a variable point on the line segment BC. Let E be the point on the circumcircle of triangle ABC lying on the opposite side of BC from A such that $\angle BAE = \angle DAC$. Let I be the incenter of triangle ABD and let J be the incenter of triangle ACE. Prove that the line IJ passes through a fixed point independent of D.*

Problem 5.6 (USA TSTST 2019 P5). *Let ABC be an acute triangle with orthocenter H and circumcircle Γ. A line through H intersects segments AB and AC at E and F, respectively. Let K be the circumcenter of $\triangle AEF$, and suppose line AK intersects Γ again at a point D. Prove that line HK and the line through D perpendicular to BC meet on Γ.*

Problem 5.7 (IMO SL 2015 G7). *Let $ABCD$ be a convex quadrilateral, and let P, Q, R, and S be points on the sides AB, BC, CD, and DA, respectively. Let the line segments PR and QS meet at O. Suppose that each of the quadrilaterals $APOS$, $BQOP$, $CROQ$, and $DSOR$ has an incircle. Prove that the lines AC, PQ, and RS are either concurrent or parallel to each other.*

Problem 5.8 (IGO 2014 P5). *Two points P and Q lying on side BC of triangle ABC and their distance from the midpoint of BC are equal. The perpendiculars from P and Q to BC intersect AC and AB at E and F, respectively. Let M be the point of intersection of PF and EQ. If H_1 and H_2 are the orthocenters of triangles BFP and CEQ, respectively, prove that $AM \perp H_1H_2$.*

Problem 5.9 (IMO SL 2012 G8). *Let ABC be a triangle with circumcircle ω and ℓ a line without common points with ω. Denote by P the foot of the perpendicular from the center of ω to ℓ. The side-lines BC, CA, AB intersect ℓ at the points X, Y, Z different from P. Prove that the circumcircles of the triangles AXP, BYP and CZP have a common point different from P or are mutually tangent at P.*